Practical No. 05

Aim: Study and implementation of TCL commands.

Software Requirements: MySQL, Oracle 10g.

Theory:

Transaction Control Language (TCL) is a critical component of **SQL** used to manage transactions and ensure **data integrity** in **relational databases**. By using TCL commands, we can control how changes to the database are committed or reverted, maintaining consistency across multiple operations.

TCL includes the following commands:

1. **BEGIN TRANSACTION:**
   - The **BEGIN TRANSACTION** command is used to explicitly begin a transaction.
   - Once started, any subsequent SQL operations such as INSERT, UPDATE, or DELETE are executed within the scope of that transaction.
   - Syntax: START TRANSACTION;

2. **SAVEPOINT:**
   - The **SAVEPOINT** command is used to **set a point within a transaction to which we can later roll back.**
   - This command allows for partial rollbacks within a transaction, providing more control over which parts of a transaction to undo.
   - Multiple savepoints can be created in a single transaction
   - **Syntax:** SAVEPOINT savepoint_name;
   - **Example:** SAVEPOINT s1;

3. **ROLLBACK:**
   - The **ROLLBACK** command is used to **undo all the transactions** that have been performed during the current transaction but have not yet been committed.
   - This command is useful for reverting the database to its previous state in case an error occurs or if the changes made are not desired.
   - **Syntax:** ROLLBACK;
ROLLBACK to savepoint_name;
   - **Example:** ROLLBACK; -- Undo entire transaction
ROLLBACK TO s1; -- Undo changes only after savepoint s1

4. **COMMIT:**

- The COMMIT command is used to **save all the transactions to the database** that have been performed during the current transaction.
- Once a transaction is committed, it becomes permanent and cannot be undone.

  This command is typically used at the end of a series of SQL statements to ensure that all changes made during the transaction are saved.
- **Syntax:** COMMIT;

Program:

```
create database new;
use new;
create table stud(ID int, name varchar(50), Branch varchar(50));
insert into stud values(1, 'Elon Musk', 'CSE');
insert into stud values(2, 'Carl Pei', 'AI');
insert into stud values(3, 'Sam Altman', 'DS');
insert into stud values(4, 'Jeff Bezos', 'IT');
start transaction;
select * from stud;
savepoint s1;
update stud set name = 'Jensen Huang' where ID = 4;
select * from stud;
savepoint s2;
insert into stud values(5, 'Larry Page', 'ETC');
delete from stud where ID = 2;
select * from stud;
savepoint s3;
rollback to s2;
select * from stud;
rollback to s1;
select * from stud;
commit;
```

Output:

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from stud;
+------+------------+--------+
| ID   | name       | Branch |
+------+------------+--------+
|    1 | Elon Musk  | CSE    |
|    2 | Carl Pei   | AI     |
|    3 | Sam Altman | DS     |
|    4 | Jeff Bezos | IT     |
+------+------------+--------+
4 rows in set (0.00 sec)

mysql> savepoint s1;
Query OK, 0 rows affected (0.00 sec)

mysql> update stud set name = 'Jensen Huang' where ID = 4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from stud;
+------+--------------+--------+
| ID   | name         | Branch |
+------+--------------+--------+
|    1 | Elon Musk    | CSE    |
|    2 | Carl Pei     | AI     |
|    3 | Sam Altman   | DS     |
|    4 | Jensen Huang | IT     |
+------+--------------+--------+
4 rows in set (0.00 sec)

mysql> savepoint s2;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into stud values(5, 'Larry Page', 'ETC');
Query OK, 1 row affected (0.00 sec)

mysql> delete from stud where ID = 2;
Query OK, 1 row affected (0.00 sec)

mysql> select * from stud;
+------+--------------+--------+
| ID   | name         | Branch |
+------+--------------+--------+
|    1 | Elon Musk    | CSE    |
|    3 | Sam Altman   | DS     |
|    4 | Jensen Huang | IT     |
|    5 | Larry Page   | ETC    |
+------+--------------+--------+
4 rows in set (0.00 sec)

mysql> savepoint s3;
Query OK, 0 rows affected (0.00 sec)

mysql> rollback to s2;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from stud;
+------+--------------+--------+
| ID   | name         | Branch |
+------+--------------+--------+
|    1 | Elon Musk    | CSE    |
|    2 | Carl Pei     | AI     |
|    3 | Sam Altman   | DS     |
|    4 | Jensen Huang | IT     |
+------+--------------+--------+
4 rows in set (0.00 sec)

mysql> rollback to s1;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from stud;
+------+------------+--------+
| ID   | name       | Branch |
+------+------------+--------+
|    1 | Elon Musk  | CSE    |
|    2 | Carl Pei   | AI     |
|    3 | Sam Altman | DS     |
|    4 | Jeff Bezos | IT     |
+------+------------+--------+
4 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> |
```

Conclusion: Hence, we have successfully studied and implemented the TCL commands.